

cs-mail: Permissioned Communication with Bonded First Contact

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Abstract

cs-mail is a communication protocol in which recipients control who may communicate with them without friction. An accepted sender communicates without paying per-message fees. A sender who has not been accepted must temporarily reserve a communication bond consisting of a processing charge and recipient collateral. Acceptance refunds every decision-open admitted bond for the directed relationship, fully cancels its unadmitted reservations, and grants standing communication permission. Rejection transfers the collateral to the recipient and the processing charge to the recipient provider. Silence returns the collateral but pays the processing charge to the provider. The design places cost on unsuccessful attempts to establish a relationship rather than on communication within an established relationship. This paper explains the problem, mechanism, incentives, identity model, privacy architecture, and path from a centralized deployment to an interoperable network. Native message content is end-to-end encrypted, while pairwise identifiers, compartmentalized services, and deliberate record expiration protect the communication graph and settlement history. Exact protocol requirements are defined separately in the *cs-mail Protocol Specification*.

Reading guide. Sections 1–3 introduce the protocol through one first contact before adding formal notation. Later sections add relationship-wide decisions, repeated attempts, privacy, incentives, settlement, and deployment. The accompanying protocol specification is normative; this paper explains the design and its tradeoffs.

1 The Communication-Permission Problem

Electronic mail allows anyone who knows an address to impose delivery, filtering, storage, and attention costs on that address. Spam filters attempt to predict whether a recipient wants each message, but they operate after the sender has already exercised an inexpensive right to transmit. Automation and generative systems continue to reduce the marginal cost of producing plausible unsolicited messages, while the recipient’s time remains scarce.

Charging for every message would address abundance by making ordinary communication expensive. That is the wrong boundary. Friends, colleagues, and organizations with an established reason to communicate should not incur a new financial toll each time they speak. The scarce event is not communication within a recognized relationship; it is an unknown sender’s attempt to establish one.

cs-mail makes that boundary explicit. Permission is a persistent, directed relationship controlled by the recipient. Known and accepted senders communicate normally. Unknown, rejected, or revoked senders may still attempt contact, but they must place value behind the attempt. Rejection declines the present solicitation rather than permanently banning the sender: a later attempt may become eligible after the applicable backoff. A separate recipient-controlled block refuses future ordinary contact until the recipient removes it.

2 A First Contact, End to End

Alice has not yet been accepted by Bob. Bob’s provider quotes a processing charge of 0.25 settlement units and collateral of 4.75 settlement units. Alice authorizes a temporary reservation of 5.00 and sends one encrypted message. The provider admits it only after the encrypted content and its delivery intent are durably recorded. The reservation does not purchase Bob’s attention or a response; admission makes the contact attempt valid under the protocol.

Bob’s relationship decision produces one of three outcomes:

Outcome	Settlement of 5.00	Future permission	Meaning
Accept	All 5.00 returns to Alice.	Alice may communicate with Bob without future ordinary bonds.	Bob grants the directed relationship.
Reject	0.25 goes to Bob’s provider; 4.75 goes to Bob.	Alice remains unaccepted and may retry only after backoff.	Bob declines this solicitation.
Ignore	0.25 goes to Bob’s provider; 4.75 returns to Alice at expiry.	Alice remains unaccepted.	No relationship decision arrived in time.

The relationship is directed: Bob accepting Alice does not mean that Alice has accepted Bob. If Bob accepts while several Alice-to-Bob bonds remain open, the decision refunds all of them atomically. If Alice remains unaccepted and tries again, the later attempt may also require backoff and an additional refundable persistence reserve. Those refinements are introduced only after the basic first-contact rule below.

3 The Core Mechanism

Let sender i attempt to contact recipient j . If j has accepted i , no bond is required. Otherwise, the destination side publishes signed contact terms and the sender reserves

$$B = C + S, \tag{1}$$

where C is a communication-processing charge and S is collateral selected by the recipient, subject to any minimum established by the recipient provider. The quote fixes the amounts, settlement unit, duration, and policy version before the sender commits funds.

The relationship may then be accepted, rejected, or left undecided until the bond expires:

$$\begin{aligned}
\text{Accept} &: C + S \rightarrow \text{sender}, \\
\text{Ignore} &: S \rightarrow \text{sender}, \quad C \rightarrow \text{recipient provider}, \\
\text{Reject} &: S \rightarrow \text{recipient}, \quad C \rightarrow \text{recipient provider}.
\end{aligned} \tag{2}$$

C and S serve different purposes. The processing charge compensates the destination service when an unsolicited attempt does not become an accepted relationship. Collateral creates a larger loss when the recipient affirmatively rejects the attempted relationship. A wanted first contact is costless apart from the temporary use of liquidity.

Reservation is not admission. The quote supplies a short admission window, and the sender or an expiry worker may cancel a still-unadmitted reservation. If the message is not admitted before that window closes, the entire $C + S + L_k$ returns to the sender, no attempt level advances, and no recipient solicitation appears. The recipient’s decision window begins at canonical admission time.

This prevents abandoned uploads from locking funds or paying a processing charge for a message the recipient provider never admitted.

4 Relationships, Not Individual Messages

Permission is directed. If j accepts i , that does not imply that i has accepted j . It is also attached to visible protocol identities: accepting one address does not automatically accept every address controlled by the same person or organization.

Most importantly, acceptance and rejection are decisions about the communicative relationship, not classifications of isolated messages. Several bonded attempts may be outstanding for the same directed identity pair. Acceptance therefore performs one atomic relationship transition. It

- establishes permission for future communication;
- refunds $C + S$ for every admitted, decision-open bond for the relationship;
- cancels and fully releases every reserved but unadmitted attempt for the relationship;
- releases active persistence reserves associated with attempts from the accepted public identity; and
- cancels the corresponding scheduled expiries.

Acceptance does not reset the repeated-attempt state shared by the sender’s private principal and this recipient identity, nor does it release reserves associated with another public identity controlled by that principal. The accepted public identity bypasses that anti-abuse state while permission remains active. Other identities do not inherit its acceptance or forgiveness.

Once the relationship is accepted, an older message cannot subsequently be rejected. Messages admitted after the transition require no bond. A concurrent reservation either linearizes before acceptance and is cancelled with a full release if still unadmitted, or linearizes afterward and is rejected as unnecessary without retaining value.

Rejection is likewise relationship-wide. It settles all bonds that are pending at the rejection’s canonical ordering point, leaves the sender outside the accepted set, and preserves the applicable repeated-attempt constraints. It declines the active solicitation, not all future contact: after the backoff, another eligible bonded attempt may open a new solicitation. The interface may present accept or reject beside a message, but the consequence applies to the directed relationship represented by that message.

A permanent contact prohibition is a different recipient decision. Blocking closes the active solicitation, settles its decision-open admitted bonds under the rejection allocation, fully cancels unadmitted reservations, and prevents new ordinary contact terms or admission until the recipient unblocks the directed identity pair. Unblocking restores eligibility for future bonded attempts; it does not accept the relationship, erase the backoff, or reopen settled bonds.

4.1 One Solicitation, Potentially Several Messages

The recipient-facing request to establish a relationship is also scoped to the directed identity pair rather than to one message. The first admitted attempt for an unaccepted relationship opens a *relationship solicitation episode* and generates one logical prompt or notification. Its presentation channel may depend on recipient settings. Later admitted attempts join that active episode. They

may add messages or update a quiet count in the grouped presentation, but they do not generate another relationship-establishment prompt.

This coalescing does not make the later attempts financially idempotent. Each is a distinct authenticated attempt with its own content, bond, deadline, and any required persistence reserve, and admission advances repeated-attempt state. Only an exact replay of the same command is idempotent. The user-facing solicitation is coalesced so that repeated messages cannot repeatedly demand a relationship decision.

Acceptance, rejection, or blocking closes the active episode and applies to the relationship and every bond open at the decision’s canonical time. Rejection allows a later eligible attempt after backoff; blocking does not. Silence is different: it is not an `IgnoreRelationship` command. The recipient simply makes no decision, and each bond expires under its own deadline. When no admitted bond in the episode remains decision-open, the episode lapses. A later eligible attempt, after the applicable backoff, may open a new episode and one new solicitation. Merely dismissing or hiding the prompt has no permission or settlement effect.

4.2 Acceptance Near Expiry

An explicit recipient decision should not lose merely because a background expiry worker happens to run at the same moment. At the same time, a settled payment cannot remain reversible indefinitely. *cs-mail* resolves this tension with a canonical event order and a bounded finalization rule.

The authoritative service durably timestamps and sequences an acceptance when it receives it. An acceptance received on or before a bond’s decision deadline takes priority over mechanical expiry. Expiry settlement is finalized only after a short processing holdback has allowed timely, already-received decisions to enter the canonical journal. The recipient receives a signed receipt for the decision.

An acceptance received after a bond’s decision deadline may still grant future communication permission and settle every other bond whose decision window remains open. It does not reopen a bond that has already expired. Thus explicit intent wins a genuine processing race, while terminal ledger settlement remains final.

5 Repeated Attempts

A sender could otherwise treat repeated losses of C as the cost of searching for eventual acceptance, particularly because silence returns S . The protocol therefore makes repeated attempts to the same unaccepted recipient progressively slower and more capital-intensive.

At attempt level k , admission may require a backoff interval Δ_k and a refundable persistence reserve L_k , with

$$\Delta_{k+1} \geq \Delta_k, \quad L_{k+1} \geq L_k. \quad (3)$$

The total reservation becomes

$$R_k = C + S + L_k. \quad (4)$$

L_k is not a fee. It is never paid to the recipient or either provider. Acceptance releases active reserves associated with attempts by the accepted public identity. It does not release reserves attached to another identity of the same private principal. Following rejection or expiry, L_k remains unavailable until its quoted release time and then returns to the sender. Cancellation before

admission releases it immediately. This raises the liquidity required for persistent contact without creating an incentive for another party to encourage failed communication.

The attempt level advances on admission rather than settlement, preventing a sender from opening many low-level attempts while the first remains unresolved. It is principal-scoped anti-abuse history and is not reset when one public identity is accepted. A reserved attempt cancelled before admission releases all funds and does not advance the level. An explicit policy may later decay old principal-level history, but acceptance of one identity is not that policy.

6 Identity, Privacy, and Security

The privacy objective of *cs-mail* extends beyond hiding the words in a message. A communication system also reveals who attempted to contact whom, whether the recipient accepted the relationship, how frequently contact was attempted, and how money moved afterward. Taken together, those facts form a relationship graph. The graph can be as sensitive as message content, because it may reveal personal, professional, medical, political, or commercial associations without revealing a single sentence of correspondence.

The initial C-SQD deployment therefore distinguishes a private principal from the public protocol identities through which that principal communicates. Alice may communicate as `alice@work.example` and `alice@personal.example`. These are two public protocol identities: they are the personas other participants can address, recognize, accept, or reject. Alice’s provider may know that both identities are controlled by the same private principal. Bob normally sees only the identity Alice uses with him, not the private principal or the fact that her two identities are connected.

The distinction serves two different purposes. Recipient permission follows the public protocol identity, because accepting Alice’s work identity should not silently accept every other identity she controls. Repeated-attempt enforcement follows the private principal with respect to a particular recipient identity, because changing addresses or operational keys should not erase the history of unwanted attempts. A privacy-preserving continuity mechanism can carry that history across a provider change without turning the principal into a public identifier.

A private principal is the provider-local subject to which control, recovery, and anti-abuse continuity are attached. It is not necessarily a verified human or legal person; depending on provider policy, it may represent a person, account, organization, or managed group. Its identifier is neither a public address nor a universal identifier shared among providers or recipients. Other domains receive only the narrowly scoped assertions they need, so acceptance of one identity does not expose or authorize the principal’s other identities.

6.1 Content and Envelope Privacy

Native message content should be end-to-end encrypted from the sender’s device to the recipient’s device. Content includes the subject, body, inline media, and attachments. Providers store and transport ciphertext and do not possess the ordinary decryption keys. The encrypted content remains outside the relationship state and financial journal, so a ledger compromise cannot become a message archive and a content-store compromise cannot rewrite settlement.

Delivery still requires a small envelope. The destination must be routable, the sender must be authenticated, and the recipient provider must be able to verify that a required bond exists. The envelope should disclose only those facts. A valid bond does not justify exposing the subject, content, legal identity, unrelated addresses, account balance, or history of other relationships.

End-to-end encryption protects content from providers and storage compromise, but it does not protect a device on which the recipient has opened the message. Device security, authenticated software updates, key recovery, key rotation, and visible warnings for unexpected key changes are therefore part of the security model.

6.2 Authenticated Declarations and Recipient Policy

Encryption need not reduce every message to an undifferentiated ciphertext blob. A small authenticated envelope can carry a coarse declared purpose, accountable origin mode, message validity time, optional payload-schema identifier, and the capability reference required for delivery. These fields let a recipient apply explicit rules without exposing the subject, body, attachments, or structured payload to the provider.

Their meaning must remain precise. A signature proves which authorized sender or gateway made a declaration; it does not prove that the ciphertext is truthfully about that purpose or that a human composed its words. A native sender signs its own declarations. A legacy gateway identifies itself as the declaration authority and cannot turn domain authentication into proof of purpose or human authorship. Recipients respond to dishonest declarations through reports, capability revocation, or blocking rather than provider inspection of native content.

Policy likewise has more than one authority. Versioned protocol policy determines quotation and settlement inputs. Provider admission policy can evaluate the authenticated envelope, permission, capability, rate, and safety evidence visible at that boundary. Recipient presentation policy controls ranking, notifications, digests, and local content processing. Only authenticated protocol commands change relationships or money. An AI system may help the recipient author or execute local rules, but its inference is not itself protocol authority.

6.3 A Private Relationship Graph

The protocol should not represent a relationship throughout the system by repeating two email addresses or by hashing those addresses in a predictable way. It should assign the directed pair an opaque, pairwise-scoped handle. A sender's relationship with one recipient receives a handle that cannot be correlated with the handle used for another recipient. The handle means nothing outside the service and relationship context for which it was created.

An accepted relationship can persist through such a handle or a recipient-issued capability. Delivery needs to know that the presented authority remains valid; it does not need a readable history of how the relationship was formed. Revocation invalidates the authority without publishing an address-book edge. Rejected, ignored, and pending attempts receive the same confidentiality as accepted ones.

Pairwise handles reduce linkability but do not make traffic analysis impossible. Timing, message size, network addresses, and distinctive settlement amounts can still connect otherwise separated events. Providers should limit retention of these signals, avoid shared tracing identifiers across privacy domains, and use batching or aggregation where it does not weaken settlement correctness.

6.4 Separation and Minimal Disclosure

No ordinary subsystem needs the complete story of a communication. The identity service needs to know that a verified principal controls a protocol identity, but not the message content. The content service needs to store encrypted bytes, but not legal identity evidence or bond ownership. The relationship service needs an opaque directed-pair handle and its current state. The ledger needs opaque account, bond, and settlement references together with amounts. The delivery service needs

a destination route and proof that the current permission or reservation rule has been satisfied. External financial rails need deposits and withdrawals, not a per-message history.

This separation is meaningful only if the cross-references are protected as carefully as the underlying records. A central operator that retains every mapping and every key can reconstruct the graph even when the individual databases look opaque. C-SQD should therefore separate access roles, encryption keys, operational logs, and retention schedules so that routine service operation never assembles a global readable relationship graph. Exceptional reconstruction should be narrow, audited, and governed by an explicit policy.

6.5 The Lifecycle of Private Records

Privacy changes as an attempt moves through the protocol. While a bond is pending, the system needs enough short-lived linkage to admit the message, prevent replay, apply backoff, receive a decision, and settle correctly. After finality and any defined dispute period, most of that event detail is no longer operationally necessary. The settlement can be reduced to the minimum accounting record that must remain, while the readable connection among identity, relationship, message, and ledger records is removed.

An accepted relationship may need to persist for years, but its history need not. The durable record can be limited to an opaque capability or pair handle, current permission state, version, and revocation information. It need not preserve every message that led to acceptance or make the two endpoints discoverable through a global index.

Encryption alone is not forgetting while the operator retains a usable key and a mapping back to people. At the end of the applicable operational, dispute, and legally required retention periods, C-SQD should delete unnecessary records and destroy the keys that protect expired identifying linkages. This cryptographic erasure must cover replicas, backups, search indexes, observability systems, and derived datasets. Anonymous aggregate measurements may remain only when they can no longer be used to reconstruct a person or relationship.

Historical linkage may also be protected by threshold encryption. For example, a sender share and a recipient share could both be required to reconstruct a private relationship archive. This is useful after the protocol no longer requires online identification of the parties. It is not a substitute for records that must remain available during settlement, fraud investigation, a dispute, or a legally required retention period. Any separately retained compliance record should contain the minimum necessary fields, use separate keys and access controls, expire under a documented schedule, and never include message content merely because a financial record must be preserved.

The system can forget only what it controls. A sender and recipient may retain their own messages and receipts, and an endpoint compromise may reveal what that endpoint knows. The protocol promise is that C-SQD and participating providers do not create an unnecessary permanent archive of everyone else's communications and relationships.

This model creates a substantial governance obligation. The identity repository, key infrastructure, recovery process, compliance archive, and access audit are sensitive systems. Their design must minimize disclosure and must also address participants for whom conventional bank-based or legal-identity verification is unavailable or inappropriate.

7 Economic Effects and Counter-Incentives

The mechanism creates four sender-side costs without taxing accepted communication:

1. ignored attempts lose C ;

2. rejected attempts lose $C + S$;
3. unsettled bonds consume liquidity; and
4. repeated attempts encounter increasing backoff and persistence reserves.

A selective sender whose contacts are commonly accepted may therefore operate at low expected cost. Indiscriminate outreach faces direct losses and working-capital requirements that scale with unsuccessful volume.

The design must also constrain recipient-side and provider-side incentives. A recipient can receive S by rejecting, and a recipient provider receives C when an attempt expires or is rejected. A mature deployment should therefore combine auditable settlement with clear pricing, prominent and easy acceptance controls, fraud detection, limits on manipulative solicitation, and dispute procedures. Providers should not be able to manufacture expiry through hidden decisions or ambiguous delivery. Signed admission and decision receipts make these behaviors observable, while governance must define permissible pricing and interface practices.

A valid bond purchases a protocol-valid opportunity to communicate. It does not purchase a recipient's attention, a favorable inbox rank, a response, or a right to bypass safety controls. Malware, unlawful content, excessive volume, and other policy violations remain independently rejectable at the transport and service layers.

8 Ledger and Settlement

The financial ledger is part of protocol validity because a bonded attempt is not valid unless the required value has been reserved. Three layers nevertheless remain distinct:

$$\boxed{\text{protocol settlement semantics} \neq \text{ledger operator} \neq \text{banking rails}} \quad (5)$$

Protocol semantics determine who owns reserved value after each event. The ledger operator maintains an append-only, auditable record of those obligations. Banks, card processors, and other regulated services fund and redeem ledger balances. Many message settlements can consequently occur internally without creating a bank transaction for each message.

Federated providers may clear those internal obligations in aggregate. For example, if Provider A owes Provider B 10,000 during a clearing period and Provider B owes Provider A 8,000, an external rail may settle the 2,000 net obligation while both providers retain the individual authenticated protocol events needed for participant receipts, audit, and dispute handling. Netting changes how providers discharge obligations; it does not change the ownership result of any individual protocol transition.

The journal should use double-entry accounting, conserve value in every transition, and make all commands idempotent. Relationship changes and their financial consequences must be atomic. A bond and a persistence reserve may each reach terminal settlement exactly once.

9 Launch Extension: Express Lanes for Scoped Communication

Permanent personal acceptance is not appropriate for every relationship. A merchant, airline, healthcare organization, public agency, or automated service may need permission to communicate only about a transaction, appointment, account, or bounded period.

Recipients may therefore issue scoped capabilities defining an authorized sender, recipient, declared-purpose and optional origin constraints, acceptable declaration authority, validity interval,

volume limit, and revocation conditions. A message whose authenticated declarations match the capability may be delivered without a bond, without claiming that the provider verified its encrypted meaning. This does not grant unrestricted standing permission. Capabilities preserve the same central principle: communication authority originates with the recipient.

Express lanes implement this capability outside the settlement kernel. A capability-authorized message neither opens nor joins a relationship solicitation nor advances repeated-attempt state. A recipient block atomically invalidates previously issued capabilities for the directed identity pair.

10 Deployment Path

An initial deployment can be centralized without collapsing the logical roles in the design. C-SQD may operate identity verification, the canonical event journal, permission state, bond ledger, delivery infrastructure, and internal clearing, while external financial providers handle funding and withdrawal.

The first implementation should focus on a deterministic protocol kernel. Given authenticated commands, explicit time, and prior state, the kernel computes the same relationship and ledger transitions without performing networking, database access, or external payments. Unit tests and property-based tests can then verify relationship-wide settlement, value conservation, replay safety, and expiry races. The surrounding services should be separated from the beginning into identity, relationship, encrypted content, ledger, and telemetry domains. Retrofitting that separation after a global identifier has entered every log and database would not recover the privacy that was already lost.

SMTP gateways can support conventional senders during migration. Later federation can allow a sender provider to reserve funds and a recipient provider to verify the reservation, deliver the message, and produce authenticated decisions. Providers may net their obligations periodically while preserving the individual protocol events. Federation requires shared representations for identities, quotes, messages, bonds, decisions, receipts, and clearing obligations, but it does not require identical internal databases or banking partners.

Operator-specific choices for identity enrollment, account funding, SMTP payment challenges, clearing, and staged launch are described separately in the *cs-mail Deployment and Migration Profile*. They are not requirements of the protocol itself.

11 Conclusion

cs-mail treats communication permission as durable protocol state and places economic friction at the first-contact boundary. Accepted communication is free of protocol message charges. Unaccepted communication carries a refundable bond whose settlement reflects whether the recipient accepts or rejects the relationship or instead makes no decision before individual bonds expire.

The central mechanism is simple, but its implementation depends on precise semantics. Acceptance and rejection apply to relationships rather than isolated messages; rejection declines the current solicitation while a distinct block prohibits future ordinary contact. Relationship decisions atomically settle all eligible admitted bonds and fully cancel unadmitted reservations. A canonical ordering rule gives timely acceptance priority over mechanical expiry without weakening finality. Repeated-attempt controls attach to private principals, while visible permission and acceptance-side reserve release remain attached to public identities; accepting one identity does not reset the principal's shared history. One active solicitation groups repeated attempts so that they cannot repeatedly demand the same relationship decision.

The same architecture treats privacy as a property of the communication graph as well as its content. Native content is end-to-end encrypted. Relationships and attempts use pairwise opaque references, and each service receives only the facts needed for its role. After settlement and the applicable retention period, the system removes unnecessary linkage and destroys expired keys rather than keeping a permanent readable history of who contacted whom.

These rules provide a path from a centralized product to an interoperable communication network. The accompanying protocol specification turns them into versioned objects, commands, state transitions, error cases, and testable invariants.